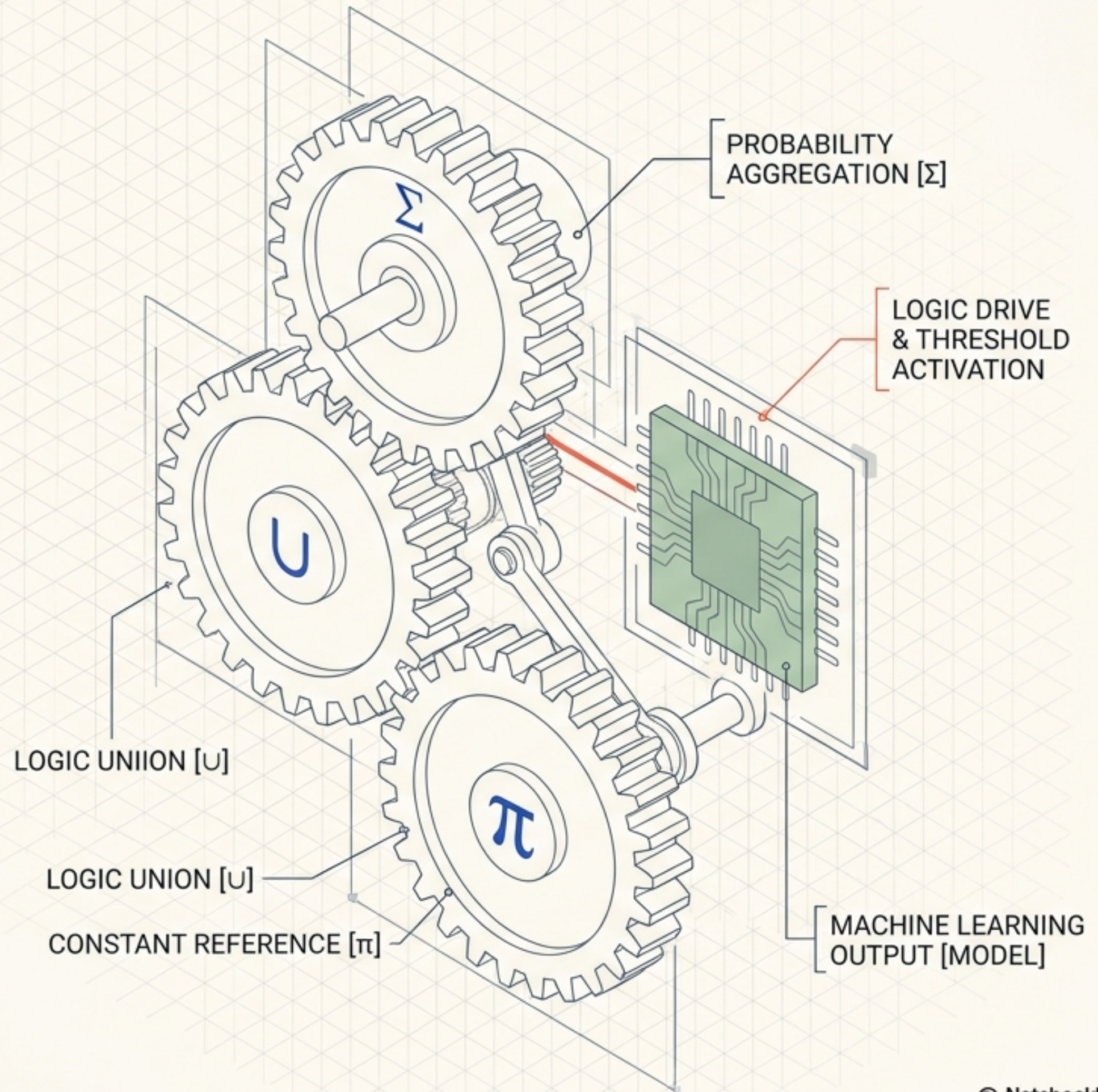
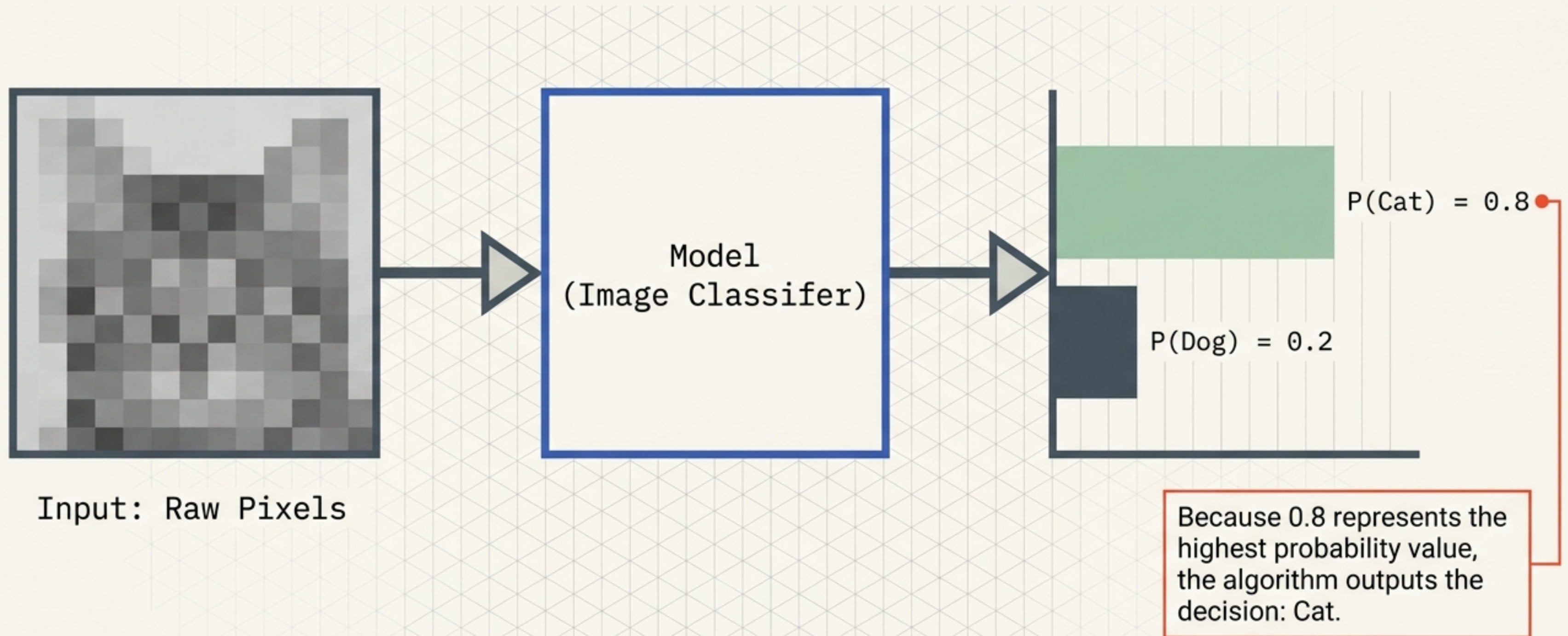


The Foundation of AI

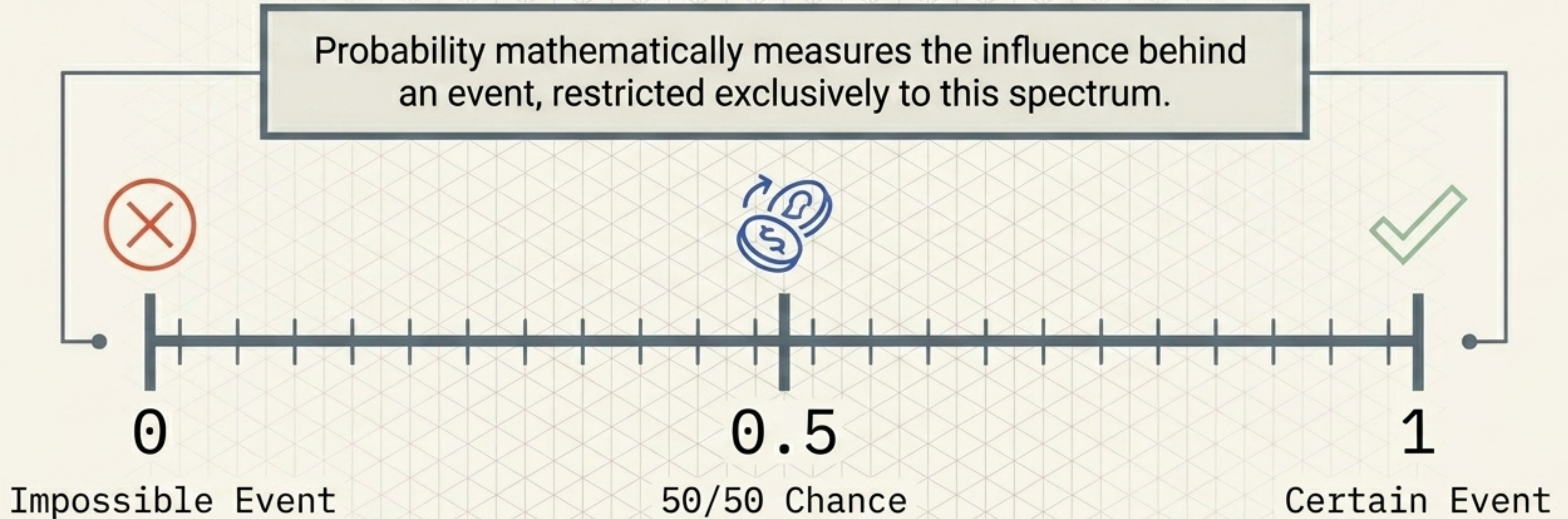
Engineering machine logic through probability.



Computers do not see objects, they calculate certainties



The fundamental engine of prediction operates strictly between zero and one



Probability measures favorable outcomes against total possibilities

$$P(\text{Event}) = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

● **Event:** The specific outcome we are targeting.

● **Sample Space (S):** The complete set of all possible results.



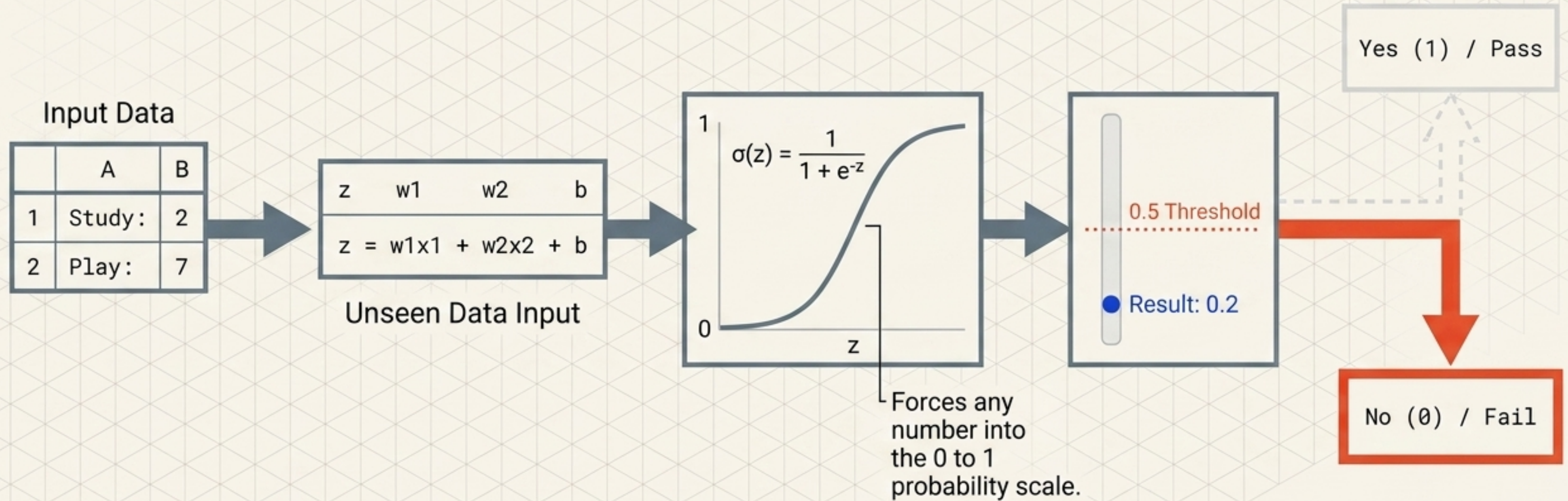
$$S = \{H, T\}$$
$$P(\text{Head}) = 1/2 = 0.5$$



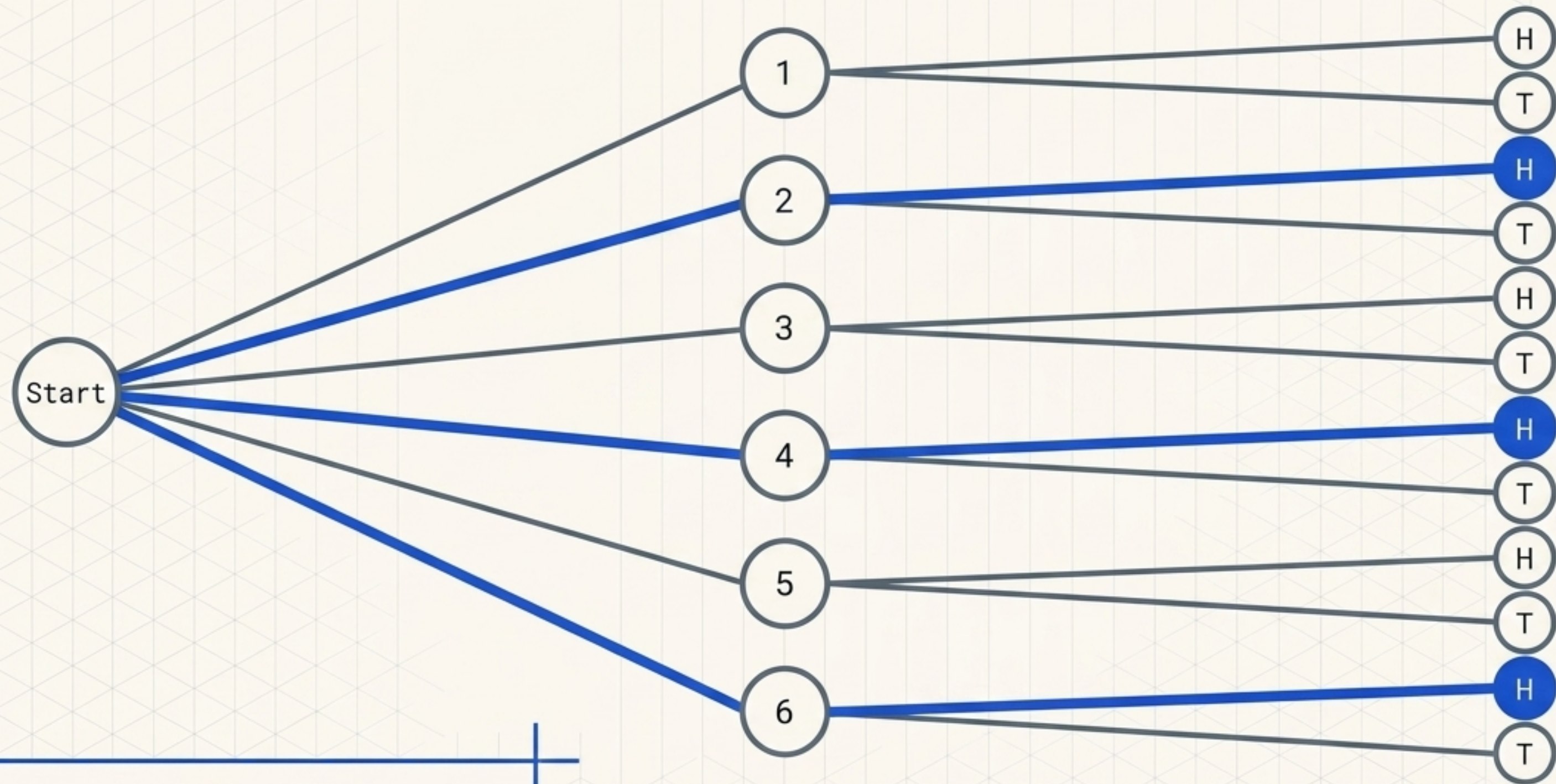
$$S = \{1, 2, 3, 4, 5, 6\}$$
$$P(4) = 1/6 \approx 0.167$$

Uniform Probability: $P(1) = P(2) = \dots = P(6)$

Logistic regression transforms raw data into binary decisions



Visualizing sequential events requires branching pathways



$$P(\text{Even, Head}) = \frac{3 \text{ Favorable}}{12 \text{ Total}} = 0.25$$

Total Sample Space $S = \{1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T\}$

Machine learning models save compute power by calculating only what isn't



$$P(\sim A) = 1 - P(A)$$

In Binary Classification, the model doesn't waste resources calculating the alternative. It simply subtracts the known probability from 1.

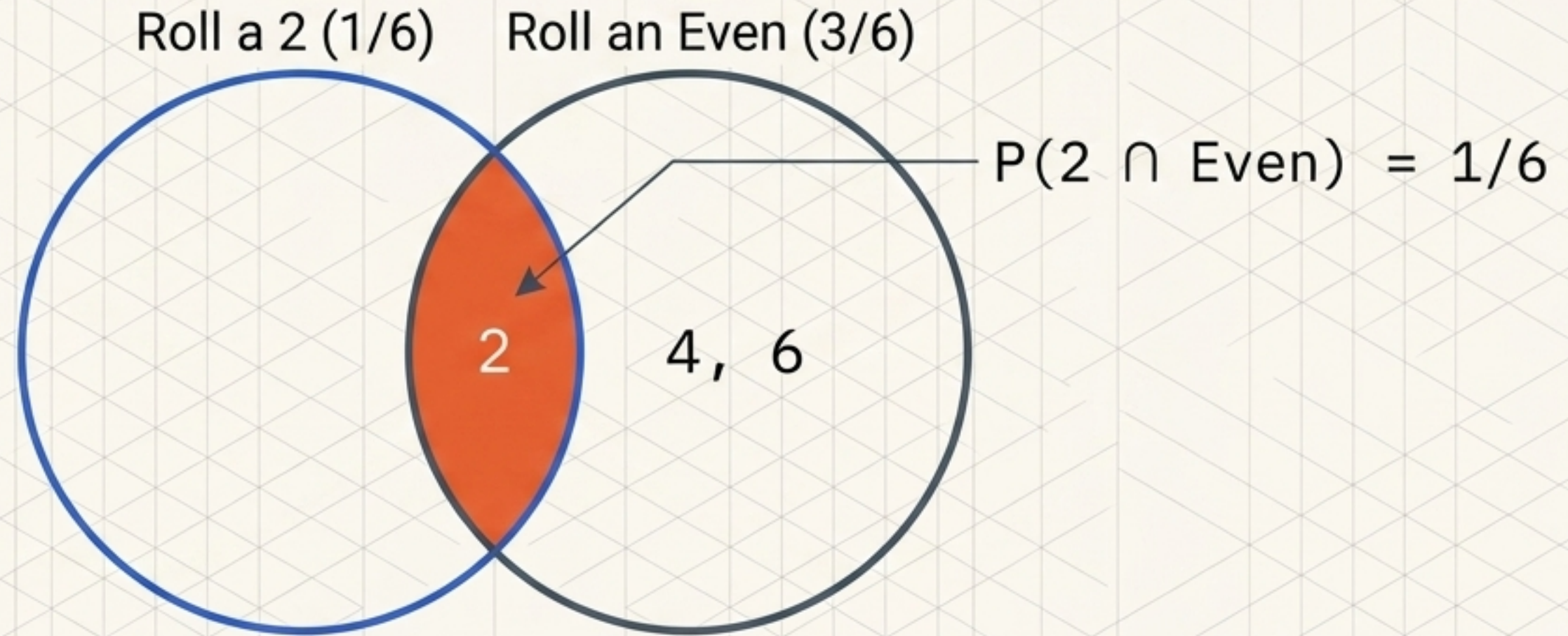
Mutually exclusive events scale linearly when combined



Rule: $P(A \cup B) = P(A) + P(B)$

Calculation: $P(2 \text{ or } 3) = 1/6 + 1/6 = 2/6 = 1/3 \approx 0.33$

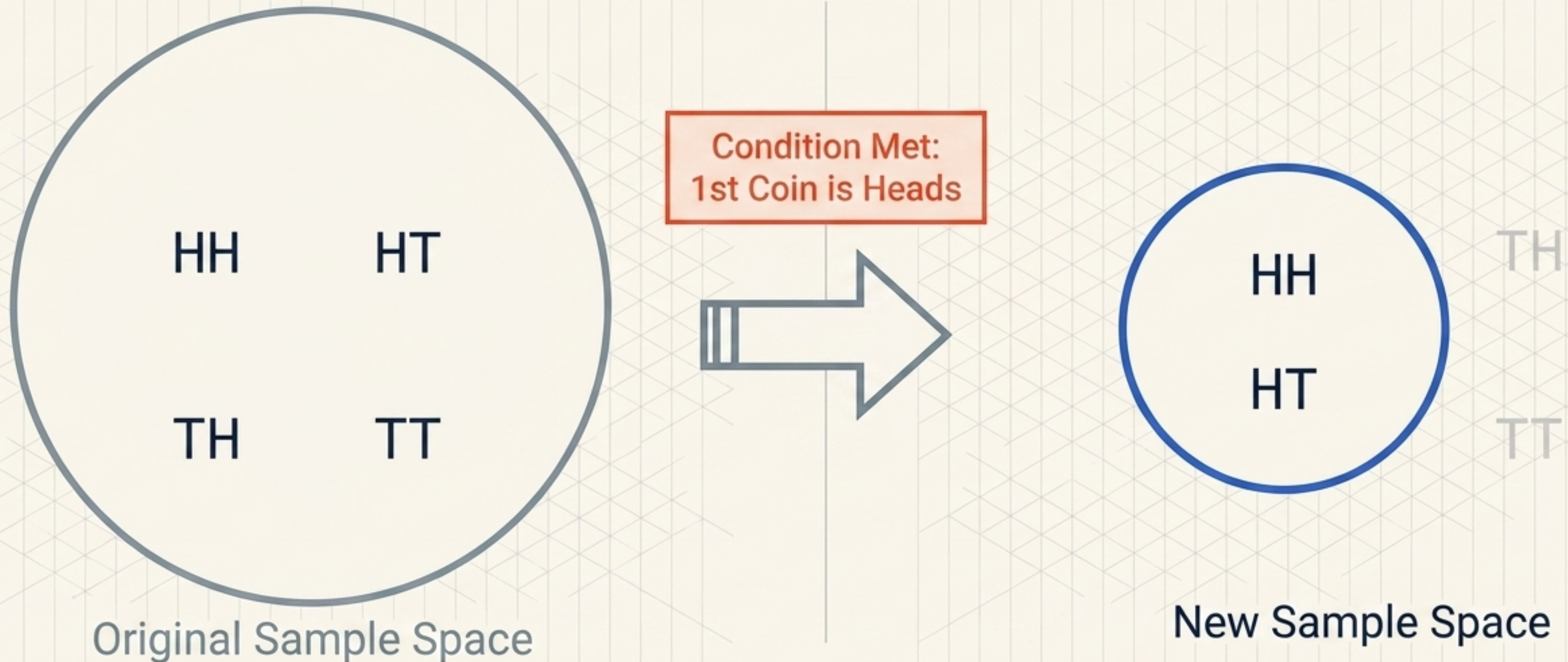
Overlapping events require subtracting the intersection to prevent double counting



General Formula: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Correction applied: $1/6 + 3/6 - 1/6 = 0.5$

Without subtracting the intersection, the element '2' would be artificially counted twice.

Known conditions physically shrink the universe of possibilities



$$P(2\text{nd is T} \mid 1\text{st is H}) = \frac{1}{2} \frac{\text{Favorable (HT)}}{\text{Total}} = 0.5$$

Translating raw data counts into the formal conditional formula

Step 1: The Logical Ratio

$$P(\text{Python} \mid \text{Male}) = \frac{\text{Male Python Lovers (30)}}{\text{Total Males (50)}} \Rightarrow \frac{n(\text{Male} \cap \text{Python})}{n(\text{Male})}$$



Step 2: Convert to Probabilities
(divide top and bottom by total sample space N=100)

$$\frac{(30 / 100)}{(50 / 100)}$$

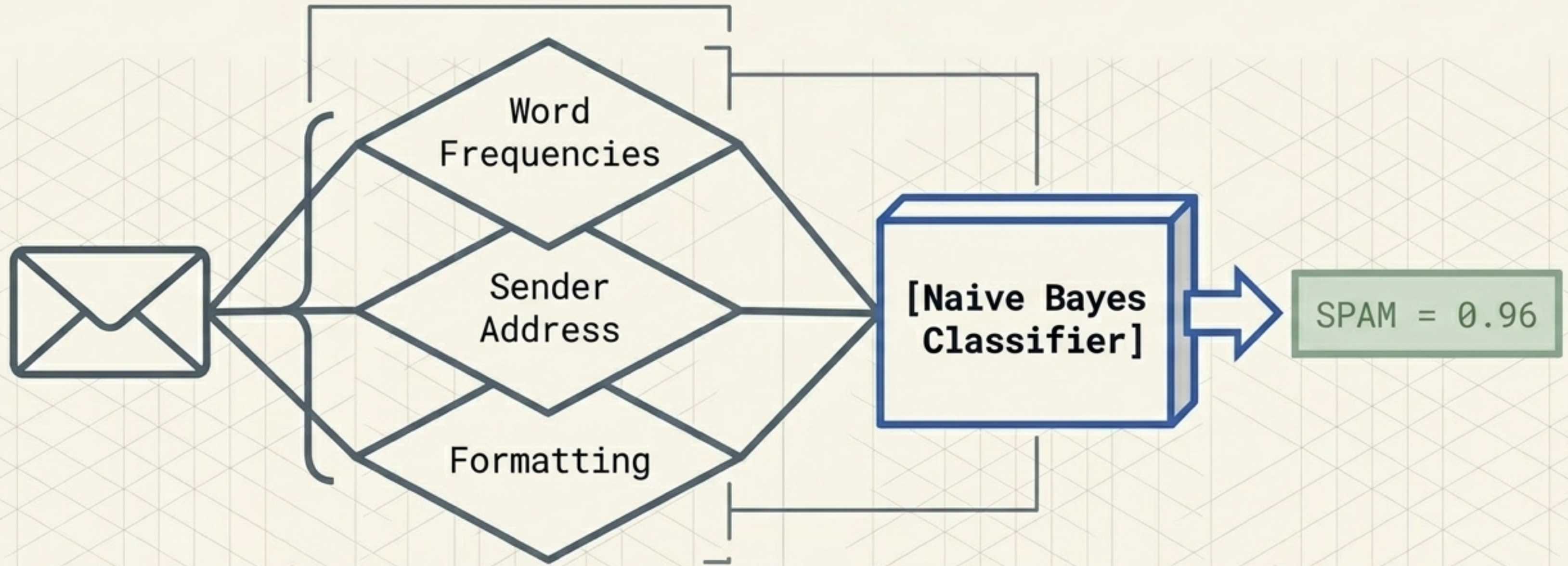


Step 3: The General Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

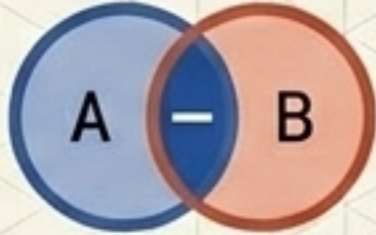
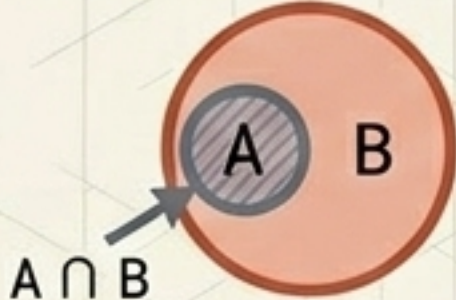
Divide the probability of both happening simultaneously by the probability of the condition that has already occurred.

Bayes' Theorem powers feature-by-feature classification



Algorithms like Naive Bayes rely entirely on Conditional Probability. By calculating the probabilities of individual features given a specific class (like evaluating word frequency given that an email is spam), the model builds a highly accurate overall classification.

The Machine Learning Blueprint: Probability Reference

Rule	Formula	Visual Concept	ML Use Case
Complement Rule	$P(\sim A) = 1 - P(A)$		Binary Classification (Logistic Regression)
Addition Rule (Disjoint)	$P(A \cup B) = P(A) + P(B)$		Combining isolated feature probabilities
Addition Rule (Joint)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$		Correcting double-counts in overlapping data
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$		Naive Bayes feature evaluation (Spam Filtering)

// end_compilation